PROBABILITY

Name: ______________________ **Date:** ________________

LEVEL 4: EXPERIMENTAL PROBABILITY

1 COIN TOSS RESULTS.

A coin was tossed 12 times.
Here are the results.

Result	Number of times
Heads	7
Tails	5

a) Which happened more often? ______________

b) What is the experimental probability of heads? ______________

c) What is the experimental probability of tails? ______________

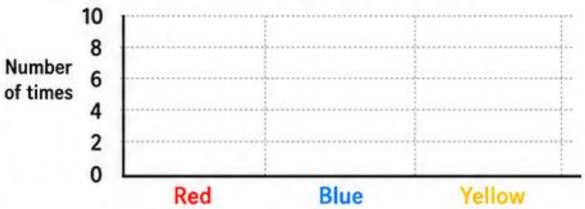
Remember: total tosses = 12

2 SPINNER TEST.

A spinner with 3 colours was spun 20 times.
Here are the results.

Colour	Number of times
Red	9
Blue	7
Yellow	4

a) Complete the bar chart using the results above.



b) Which colour occurred most? ______________

c) Which colour occurred least? ______________

d) Experimental probability of red = ______________

e) Experimental probability of blue = ______________

f) Experimental probability of yellow = ______________

Remember: total spins = 20

3 BAG EXPERIMENT.

A bag of counters was picked from 15 times. Here are the results.

Shape	Number of times
Star	6
Heart	5
Circle	4

a) Which shape seems most likely to be picked?

____

b) Which shape seems least likely to be picked?

____

c) Experimental probability of star = ______________

d) Experimental probability of heart = ______________

e) Experimental probability of circle = ______________

Remember: total picks = 15

4 COMPARE THEORETICAL AND EXPERIMENTAL.

For a fair coin, the theoretical probability of heads is $\frac{1}{2}$ (one half).

In Activity 1, the coin was tossed 12 times and there were 7 heads.

a) Is the experimental probability of heads ($\frac{7}{12}$) exactly the same as $\frac{1}{2}$? ☐ Yes ☐ No

b) Is $\frac{7}{12}$ just close to $\frac{1}{2}$? ☐ Yes ☐ No

c) **Challenge!** Explain your answer.

____

____

Great job! You are becoming a probability superstar!

Name: .... Date: ....

probability - fractions

level 3

1 spinner probability

P(red) = ☐

P(blue) = ☐

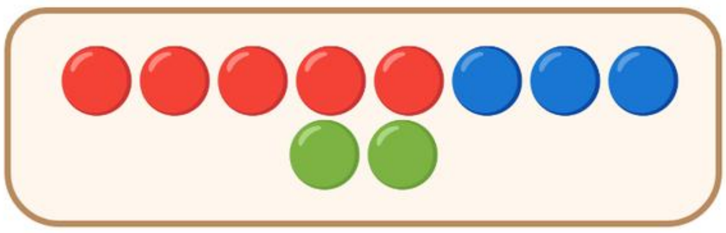
P(green) = ☐

P(yellow) = ☐

The spinner has 8 equal sections. Write each probability as a fraction.

2 marbles in a bag

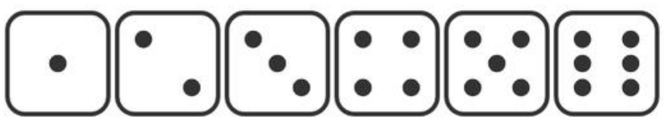
Write each probability as a fraction.



10 marbles in the bag.

P(red) = ☐ P(blue) = ☐ P(green) = ☐

3 rolling a die



A fair die is rolled. Write each probability as a fraction (out of 6).

P(rolling a **4**) = ☐ P(an **even** number) = ☐ P(a number **more than 4**) = ☐

Name: ______________________ Date: ______________________

PROBABILITY

LEVEL 1: LIKELY / UNLIKELY

1 Circle likely or unlikely.

1. 
See the sun in daytime.
likely unlikely

2. 
See a fish fly.
likely unlikely

3. 
Pick a red apple from the bowl.
likely unlikely

4. 
See snow in a desert.
likely unlikely

5. 
Roll a number from 1 to 6 on a die.
likely unlikely

6. 
Get a cat from a bag of only dog pictures.
likely unlikely

2 Sort the events.

Draw a line from each card to the correct box.

IMPOSSIBLE	UNLIKELY	LIKELY


The sun comes out at night.


It rains on a rainy day.


Pick a donut from a box of donuts.


Find two left socks in a drawer.


Your birthday is in this year.


See a snowman in summer.

3 Color the boxes.

Color the box **GREEN** if the event is likely. Color the box **ORANGE** if the event is unlikely.

1. It will rain tomorrow.

☐

2. You win a prize at school.

☐

3. You find money on the ground.

☐

4. You see a dinosaur in the park.

☐

5. You eat breakfast today.

☐

6. You jump to the moon today.

☐

7. It snows in summer where you live.

☐

8. You draw a heart in class today.

☐

4 Challenge!

Which is more likely:
picking a blue marble from a bag with 5 blue and 1 yellow, or picking a yellow marble?

My answer: ______________________

Why? ______________________

____

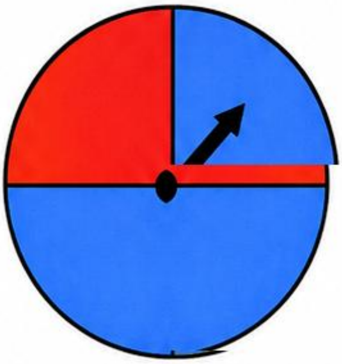
Name: ______________________ Date: ______________

PROBABILITY

- LEVEL 2: SIMPLE PROBABILITY -

1 LOOK AT THE SPINNER.

The spinner has 4 equal sections.



a) What color is most likely?

____

b) What color is least likely?

____

c) Is red certain, likely, unlikely, or impossible?

____

d) Is green impossible?

2 ROLL THE DIE.

A fair six-sided die is rolled.

a) What is the probability of rolling a 3? [ ] out of 6

b) What is the probability of rolling an even number? [ ] out of 6

c) What is the probability of rolling a number greater than 4? [ ] out of 6

3 PICK FROM THE BAG.

Look at the shapes in the bag.

a) Which shape is most likely?

____

b) Which shape is least likely?

____

c) How many possible shapes are there?

____

d) What is the chance of picking a heart?

[ ] out of 10

4 MATCH THE EVENT.

Draw a line to match each event to the correct word.

1  The sun will rise tomorrow.

2  You will find a clover in your math book.

3  You will catch a fish when you go swimming.

4  You roll a 6 on a six-sided die.

5  It will snow on a hot summer day.

6  You will wear shoes to school.

• **CERTAIN** (Always happens)

• **LIKELY** (Usually happens)

• **UNLIKELY** (Does not usually happen)

• **IMPOSSIBLE** (Cannot happen)